

DEAM Unit II – Theory Questions and Answers

1. 2:1 Multiplexer - Inputs: I0, I1; Select line: S; Output: Y - Function: Y = I0 when S=0, Y = I1 when S=1 - Logic Expression: $Y = S'I0 + SI1$ - Applications: Data routing, digital circuits

2. 4:1 Multiplexer - Inputs: I0–I3; Select lines: S0, S1; Output: Y - Logic Expression: $Y = I0S1'S0' + I1S1'S0 + I2S1S0' + I3S1S0$ - Applications: CPU operations, data selection

3. 8:1 Multiplexer - Inputs: I0–I7; Select lines: S0–S2; Output: Y - Logic Expression: $Y = I0S2'S1'S0' + \dots + I7S2S1S0$ - Applications: Memory address decoding, data routing

4. 16:1 Multiplexer - Inputs: I0–I15; Select lines: S0–S3; Output: Y - Applications: Large data selection, microprocessor interfacing

5. Full Subtractor - Inputs: A, B, Bin; Outputs: Difference D, Borrow_out Bout - Logic Expressions: $D = A \oplus B \oplus Bin$, $Bout = A'B + A'Bin + BBin$ - Applications: Arithmetic operations

6. Half Adder - Inputs: A, B; Outputs: Sum S, Carry C - Logic Expressions: $S = A \oplus B$, $C = A \cdot B$ - Truth Table:

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

 - Applications: Basic arithmetic circuits

7. Full Adder - Inputs: A, B, Cin; Outputs: Sum S, Carry Cout - Logic Expressions: $S = A \oplus B \oplus Cin$, $Cout = AB + BCin + ACin$ - Truth Table:

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

 - Applications: Multi-bit addition

8. Half Subtractor - Inputs: A, B; Outputs: Difference D, Borrow B - Logic Expressions: $D = A \oplus B$, $B = A'B$ - Truth Table:

A	B	D	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

9. 2:4 Decoder - Inputs: A, B; Outputs: Y0–Y3 - Logic Expressions: $Y0 = A'B'$, $Y1 = A'B$, $Y2 = AB'$, $Y3 = AB$ - Applications: Memory decoding, demultiplexing

10. Parity Generator and Checker - Parity Generator: Adds parity bit (even/odd) to data - Parity Checker: Detects error if parity mismatch - Applications: Error detection in digital communication

11. Types of Multiplexer and Advantages - Types: 2:1, 4:1, 8:1, 16:1 - Advantages: - Reduces number of wires and gates - Simplifies circuit design - Selects one input from many sources

12. Full Adder Using Decoder - Functions: $S(x,y,z) = \Sigma(1,2,4,7)$, $C(x,y,z) = \Sigma(3,5,6,7)$ - Use 3-to-8 decoder to generate minterms - OR appropriate outputs for Sum and Carry

13. Difference Between MUX, DEMUX, Decoder - MUX: Many inputs $\rightarrow$ 1 output - DEMUX: 1 input $\rightarrow$ Many outputs - Decoder: n inputs $\rightarrow 2^n$ outputs representing input combinations

14. Design 16:1 MUX Using 4:1 MUX (Cascading) - Use four 4:1 MUXs for lower selection - Fifth 4:1 MUX combines outputs for final result - Select lines S0-S3 control hierarchy

15. Parity Generator and Checker - Reiterated from Q10; ensure understanding of parity bit generation and error checking.